Web Technologies

Basics of HTML

01.What does HTML stand for?
HyperText Markup Language.

02.What is the purpose of HTML?
HTML structures web content and tells browsers how to display text, images, links, etc.

03.What is a tag in HTML?
Tags are the elements enclosed in angle brackets (e.g., `<p>`) that define HTML elements.

04. What is the function of the `<html>` tag?
It wraps the entire HTML document and signifies the beginning and end of an HTML page.

05. What is the role of the `<head>` tag?
It contains meta-information about the document, like the title, charset, and linked resources.

06.What does the `<title>` tag do?
Sets the title of the webpage, shown on the browser tab.

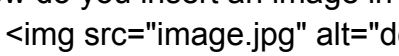
07.Where does the content of a web page go?
Inside the `<body>` tag.

08.How do you insert a line break in HTML?
Use the `<br>` tag.

09.What tag is used to create a paragraph?
`<p>` tag

10. What tag creates a heading in HTML?
`<h1>` to `<h6>`, where `<h1>` is the largest.

Media & Links

01. How do you insert an image in HTML?


02.What tag is used to create a hyperlink?

<a href="url">Link</a>

03.What does the `alt` attribute in images do?

Provides alternative text for screen readers and shows if the image can't load.

04. How do you make a link open in a new tab?

Add `target="_blank"` to the `<a>` tag.

05.What is the purpose of the `href` attribute?

It specifies the destination URL of a link.

Lists & Tables

01. What tags are used to create an unordered list?

`<ul>` for the list, and `<li>` for list items.

02.What tag creates an ordered list?

`<ol>`, used with `<li>` items.

03.How do you create a table in HTML?

Use `<table>`, with `<tr>`, `<th>`, and `<td>` tags.

04.What tag defines a table row?

<tr>

05. How do you merge table columns?

Use the `colspan` attribute in `<td>` or `<th>`.

06. How do you merge table rows?

Use the `rowspan` attribute in `<td>` or `<th>`.

HTML forms

01.What tag is used to create a form?

<form>

02.How do you create a text input in a form?

<input type="text">

03.What does the `<label>` tag do?

Defines a label for a form input.

04.How do you create a submit button?

`<input type="submit">` or `<button type="submit">`

05.What is the `action` attribute in a form for?

Specifies where to send the form data.

06.What does `method="post"` do in a form?

Sends form data in the body of the HTTP request (more secure than GET).

07. How do you create a checkbox?

<input type="checkbox">

08.How do you create a dropdown menu?

Use `<select>` with `<option>` tags.

09. What input type creates a radio button?

<input type="radio">

10. What does the `name` attribute in form elements do?

It assigns a name to the form data for server-side reference.

HTML attributes & semantics

01.What is an attribute in HTML?

Extra information inside a tag that modifies behavior or appearance (e.g., `class`, `id`, `style`).

02.What does the `id` attribute do?

Uniquely identifies an HTML element.

03.What is the `class` attribute used for?

Assigns one or more class names for CSS or JavaScript reference.

04. What is semantic HTML?

HTML that clearly describes its meaning (e.g., `<article>`, `<section>`, `<header>`).

05.What is the `<nav>` element used for?

Defines navigation links.

06.What does the `<footer>` tag represent?

Defines the footer section of a document or section.

07.What is the `<section>` tag for?

Groups related content within a document.

08.What is the difference between `<div>` and `<span>`?

`<div>` is a block-level container; `<span>` is inline.

09.What is the `<article>` tag used for?

Represents self-contained content like a blog post.

10.What is the `<aside>` tag used for?

Defines content indirectly related to the main content (like sidebars).

Advanced HTML

01.What does the `<!DOCTYPE html>` declaration do?

Declares the document type and version of HTML.

02. How do you comment in HTML?

<!-- This is a comment →

03.Can HTML be case-sensitive?

HTML is not case-sensitive, but lowercase is standard.

04.What is the purpose of the `<meta>` tag?

Provides metadata like charset, author, and viewport settings.

05.What is the `charset="UTF-8"` declaration for?

Specifies the character encoding of the document.

06. What is the difference between block-level and inline elements?

Block-level starts on a new line; inline does not.

07.Can you nest HTML elements?

Yes, as long as they are properly closed and follow HTML syntax.

08.What is the use of `<script>` in HTML?
Embeds or links to JavaScript.

09. What tag is used to link CSS to HTML?
<link rel="stylesheet" href="style.css">

10.What is the use of the `<canvas>` tag?
Provides a drawable region for dynamic graphics via JavaScript.

History

01.Who is the inventor of the World Wide Web?
Tim Berners-Lee.

02. In what year was the World Wide Web invented?
1989.

03. What was the first web browser called?
WorldWideWeb (later renamed Nexus).

04.What organization did Tim Berners-Lee work for when he invented the web?
CERN (European Organization for Nuclear Research).

05.When was the first website launched?
1991.

06.What was the URL of the first website?
http://info.cern.ch

07.What is the significance of HTML in web history?
HTML is the standard markup language used to create and structure web pages.

08.What year was HTML 2.0 released?
1995.

09.Who developed HTML?
Tim Berners-Lee.

10.What is the W3C?
World Wide Web Consortium, the main international standards organization for the web.(1994)

11.What organization originally managed HTML standardization before W3C?
Internet Engineering Task Force (IETF).

12. What major version introduced semantic elements like `<article>` and `<section>`?
HTML5.

13.When was HTML5 officially finalized as a standard?
2014.

14.What replaced HTML 4.01?
HTML5.

15.What major shift did HTML5 represent?
A move toward rich, interactive applications without the need for plugins like Flash.

16.What was the purpose of XHTML?
To combine the rigor of XML with the flexibility of HTML.

17.What is the difference between HTML and HTTP?
HTML structures web content, while HTTP is the protocol for transferring data on the web.

18. What was the Netscape browser's role in web history?
It helped popularize the web in the mid-1990s.

19.What is the "browser wars"?
A competition between Netscape and Internet Explorer during the 1990s for dominance in web browsing.

20. What browser was bundled with Windows and led to the downfall of Netscape?
Internet Explorer.

21.When was CSS introduced to separate content from design?
1996.

22. What is JavaScript and when was it introduced?
A scripting language for web interaction, introduced in 1995 by Netscape.

23. What was the purpose of the DOM (Document Object Model)?
To allow scripts to dynamically access and update content, structure, and style of documents.

24. When did mobile web usage start to rise significantly?
Late 2000s, especially after the iPhone's release in 2007.

25.What is the significance of the `<canvas>` element introduced in HTML5?
It allows for dynamic, scriptable rendering of 2D shapes and images.

26.What is the Internet Archive's "Wayback Machine"?
A digital archive of the World Wide Web dating back to 1996.

27.What does URL stand for?
Uniform Resource Locator.

28.What does URI stand for?
Uniform Resource Identifier.

29.What was the Mosaic browser?
One of the first widely-used graphical web browsers, released in 1993.

30.What role did Apache play in the history of the web?
It became the dominant web server software, enabling millions of websites.

CSS Basics

01.What does CSS stand for?
Cascading Style Sheets.

02.What is the purpose of CSS?
It styles HTML elements (layout, colors, fonts, etc.).

03.How do you include CSS in a web page?
Inline, Internal (`<style>`), or External (`<link rel="stylesheet">`).

04.What is the correct syntax for a CSS rule?
selector { property: value; }

05. How do you select an element by class in CSS?
.classname { }

06. How do you select an element by ID in CSS?
#idname { }

07.What symbol is used to select child elements?

`>` (e.g., `div > p`)

08. How do you comment in CSS?

/* comment */

09.What is a CSS selector?

A pattern used to select HTML elements for styling.

10. What is the default position value of an HTML element?

Static

Properties & values

01.How do you change the text color in CSS?

color: red;

02. How do you set the background color?

background-color: yellow;

03. How do you set a font family?

font-family: Arial, sans-serif;

04.How do you set font size?

font-size: 16px;

05.What unit is relative to the parent element's font size?

em

06. What unit is relative to the root element?

rem

07. How do you make text bold?

font-weight: bold;

08.How do you center text?

text-align: center;

09.How do you underline text?

text-decoration: underline;

10.What is the difference between `margin` and `padding`?

Margin is outside the element, padding is inside the border.

Box Model

01.What are the four components of the box model?

Content, Padding, Border, Margin.

02.How do you set a border?

border: 1px solid black;

03.How do you set different margins on each side?

`margin-top`, `margin-right`, etc.

04.What does `box-sizing: border-box;` do?

Includes padding and border in the element's total width and height.

05.How do you hide an element?

`display: none;` or `visibility: hidden;`

Positioning & Display

01.What are the values of the `position` property?

`static`, `relative`, `absolute`, `fixed`, `sticky`

02.What does `position: absolute` do?

Positions the element relative to the nearest positioned ancestor.

03.What does `z-index` control?

The stack order of overlapping elements.

04.What is the difference between `inline` and `block`?

`inline` doesn't break lines; `block` does.

05.What is `display: flex;` used for?

To create a flexible layout model for arranging items.

Flexbox & grid

01.What is the main axis in flexbox?
The direction set by `flex-direction`.

02.How do you center items in flexbox?
justify-content: center; align-items: center;

03.What does `flex-wrap` do?
Allows items to wrap onto multiple lines.

04.What does `grid-template-columns` define?
Column structure in a grid layout.

05.What is the difference between flexbox and grid?
Flexbox is one-dimensional; grid is two-dimensional.

Advanced CSS

01.What does `:hover` do in CSS?
Applies styles when the mouse hovers over an element.

02.What are pseudo-elements?
They style specific parts of elements (e.g., `::before`, `::after`).

03.What does `!important` do?
Forces a rule to override other rules.

04. What is a media query?
CSS rule for responsive design based on screen size or device.

05. What is the syntax for a media query?
@media (max-width: 600px) { ... }

06.What does `opacity` control?
The transparency of an element.

07. How do you make an element responsive?
Use percentages, `max-width`, and media queries.

08.What is the difference between `visibility: hidden` and `display: none`?
`hidden` hides but keeps space; `none` removes it from the layout.

09.What property makes text overflow with ellipsis?

`text-overflow: ellipsis;`

10.What is a transition in CSS?

A smooth change from one style to another.

11. How do you animate in CSS?

Using `@keyframes` and `animation` properties.

12. What is the difference between `relative` and `absolute` positioning?

`relative` moves from its normal position; `absolute` is positioned to the nearest ancestor.

13.What is the `calc()` function?

It allows dynamic calculations in CSS (e.g., `width: calc(100% - 50px)`).

14.What does `inherit` do?

What does `inherit` do?

15.What does `initial` do?

Resets a property to its default value.

Bootstrap Basics

01.What is Bootstrap?

A popular front-end framework for building responsive, mobile-first websites.

02.Who developed Bootstrap?

Originally developed by Twitter.

03.What is the latest stable version of Bootstrap?

(As of 2024: Bootstrap 5.3) – Check official site for updates.

04. What is the CDN link for Bootstrap?

A Content Delivery Network link to include Bootstrap without downloading files.

05.What are the main components of Bootstrap?

Layout, Content, Components, Utilities, Helpers.

06.What are the key files required to use Bootstrap?
CSS and JavaScript files (or their CDN links).

07. What does mobile-first mean in Bootstrap?
Designs are optimized for smaller screens first, then scale up.

08.How do you include Bootstrap via CDN? (Content Delivery Network)
Add `<link>` and `<script>` tags in your HTML `<head>` and before `</body>`.

Grid System & Layout

01.What is the Bootstrap grid system based on?
A 12-column layout.

02.What class defines a row in Bootstrap?
.row

03. What class defines a column in Bootstrap?
`.col` or `.col-sm-4`, etc.

04.What is the use of `.container` class?
It wraps content and provides fixed-width layout.

05.What is `.container-fluid`?
A full-width container that spans the entire screen.

06.What class adds responsive breakpoints?
`.col-sm-*`, `.col-md-*`, `.col-lg-*`, `.col-xl-*`

07.How do you center content in Bootstrap?
Use utility classes like `mx-auto`, `text-center`.

08.What is the purpose of gutters in Bootstrap?
Provides spacing between columns.

09.How do you offset columns in Bootstrap?
Using `.offset-*` classes.

Utilities

01.What are Bootstrap utility classes?

Classes that apply a single, specific style (e.g., `m-3`, `text-danger`).

02. How do you apply margin in Bootstrap?

Use `m-*`, `mt-*`, `mb-*`, etc.

03.How do you apply padding in Bootstrap?

Use `p-*`, `pt-*`, `pb-*`, etc.

04.What does `d-none` do?

Hides the element.

05.What is the class for making text bold?

Fw-bold

06.How do you make an element visible only on large screens?

Use `d-none d-lg-block`

07.What does `.text-muted` do?

Makes the text appear faded or gray.

Components

01.How do you create a Bootstrap button?

<button class="btn btn-primary">Click</button>

02.What class creates a large button?

Btn-lg

03.How do you create an alert?

Success!

04. How do you dismiss an alert?

Add `alert-dismissible` and a close button.

05.What is a card in Bootstrap?

A flexible content container with header, body, and footer.

06.How do you add images in a card?

Use `<img class="card-img-top">`

07.What class is used for a navigation bar?

.navbar

08.How do you make the navbar responsive?

Use `.navbar-expand-*` and `.collapse`

09.What is a Bootstrap modal?

A pop-up dialog box.

10.What class is used to toggle a modal?

`data-bs-toggle="modal"` and `data-bs-target="#id"`

11.What is a tooltip in Bootstrap?

A small pop-up label shown on hover.

12.How do you enable tooltips?

Initialize via JavaScript: `new bootstrap.Tooltip(...)`

Forms & Inputs

01.How do you create a Bootstrap form?

Use `<form>` with `.form-control`, `.form-group` etc.

02.What class styles form controls?

.form-control

03.How do you group form elements inline?

Use `.form-inline`

04. What class adds validation styles?

`.is-valid` or `.is-invalid`

Responsive & helpers

01.What does `.img-fluid` do?

Makes images scale with the parent container.

02. What does `.w-100` do?
Sets width to 100%.

03.How do you align items using flex in Bootstrap?
Use classes like `d-flex`, `justify-content-center`.

04.How do you make a layout responsive?
Use responsive breakpoints: `sm`, `md`, `lg`, `xl`, `xxl`.

05.What class makes text responsive?
Bootstrap doesn't resize text by default; use media queries or `fs-*` classes.

06. How can you customize Bootstrap?
By overriding classes in your CSS or using Sass variables.

JS Basics

01.What is JavaScript?
A programming language used to create dynamic and interactive content on web pages.

02.Is JavaScript the same as Java?
No. Despite the similar name, they are different languages.

03. How do you write a comment in JavaScript?
`// Single line`, `/* Multi-line */`

04.How do you declare a variable in JavaScript?
Using `var`, `let`, or `const`.

05.What is the difference between `let` and `const`?
`let` allows reassignment, `const` does not.

06.What are JavaScript data types?
String, Number, Boolean, Undefined, Null, Object, Symbol, BigInt.

07.What is `typeof` used for?
To check the data type of a variable.

08.What is the result of `typeof null`?

`"object"` (a known bug in JS)

09.What is NaN in JavaScript?

Stands for "Not-a-Number".

10. How do you convert a string to a number?

`Number("123")` or `parseInt("123")`

Operators & control flow

01.What is the difference between `==` and `===`?

`==` compares value, `===` compares value and type.

02. What does `||` do in JavaScript?

Logical OR.

03.What is a ternary operator?

condition ? valueIfTrue : valueIfFalse

04.What is a switch statement?

A control statement for multiple conditions.

05.What are falsy values in JS?

`false`, `0`, `""`, `null`, `undefined`, `NaN`

Loops & Functions

01. Name JavaScript loop types.

`for`, `while`, `do...while`, `for...in`, `for...of`.

02.What is the difference between `for...in` and `for...of`?

`for...in` iterates over object keys, `for...of` iterates over values in iterable objects.

03.How do you define a function?

function myFunc() { }

04.What is a callback function?

A function passed as an argument to another function.

05.What is an arrow function?

A shorter syntax for function expressions: `() => { }`

Array & Objects

01.How do you declare an array?

let arr = [1, 2, 3];

02.How do you access the first element in an array?

Arr[0]

03.What is the use of `push()`?

Adds an element to the end of an array.

04.What does `pop()` do?

Removes the last element from an array.

05.How do you loop through an array?

`for`, `forEach`, `map`, etc.

06.How do you create an object?

let obj = { key: "value" };

07.How do you access an object's property?

`obj.key` or `obj["key"]`

08.How do you delete a property from an object?

delete obj.key

DOM Manipulation

01.What does DOM stand for?

Document Object Model.

02.How do you select an element by ID?

document.getElementById("id")

03.How do you select elements by class?

document.getElementsByClassName("class")

04.How do you change the text of an element?
element.textContent = "new text"

05.How do you change the HTML of an element?
element.innerHTML = "bold"

06. How do you change CSS with JavaScript?
element.style.color = "red"

07.How do you create a new HTML element in JS?
document.createElement("div")

08.How do you append a new element to the DOM?
parent.appendChild(child)

Events

01.How do you add a click event?
element.addEventListener("click", function)

02. What is `event.preventDefault()` used for?
Stops default browser action.

03.What is `event.target`?
The element that triggered the event.

04.How do you handle form submissions?
Add `submit` event listener to the form.

ES6 and Beyond

01.What is destructuring?
Extracting values from arrays or properties from objects.

02. What are template literals?
Strings using backticks and `${}` for interpolation.

03.What are promises?
Represent asynchronous operations.

04.What is async/await?
A syntax to handle promises more cleanly.

05. What is the spread operator?

`...` to spread array/object values.

06.What is a class in JS?

A blueprint for creating objects (introduced in ES6).

07.What is a constructor?

A function that initializes new object instances.

08.What is `this` keyword?

Refers to the current object context.

09.What is closure in JavaScript?

A function remembering its lexical scope even when executed outside.

10.What is hoisting?

JS moves declarations to the top of scope during compile.

JQuery Basics

01.What is jQuery?

A fast, small, and feature-rich JavaScript library that simplifies HTML DOM manipulation, event handling, and AJAX.

02.How do you include jQuery in an HTML page?

By adding a `<script>` tag with a CDN link or a local jQuery file.

03.What does `$` represent in jQuery?

It is a shorthand alias for the `jQuery` function.

04.What is the purpose of `$(document).ready()`?

To run code after the DOM is fully loaded.

05. How is jQuery different from vanilla JavaScript?

jQuery simplifies DOM manipulation, cross-browser issues, and event handling.

Selectors

01. How do you select all paragraphs in jQuery?

$("p")

02. How do you select an element with a specific id?
$("#id")

03.How do you select elements by class?
$(".class")

04.How do you select all elements of a certain type with a specific class?
$("element.class")

05.How to select the first element of a set?
$("selector").first()

Events

01.How do you attach a click event handler in jQuery?
$("selector").click(function() { ... });

02.What is the `.on()` method used for?
To attach event handlers for one or more events to elements.

03.How do you remove an event handler?
Using `.off()` method.

04.How do you handle multiple events at once?
$("selector").on("click mouseenter", function() { ... });

05. How do you prevent the default action of an event?
Using `event.preventDefault();`

DOM Manipulation

01.How do you change the text of an element?
$("selector").text("new text")

02. How do you change the HTML inside an element?
$("selector").html("new HTML")

03.How do you add a class to an element?
$("selector").addClass("className")

04.How do you remove a class from an element?
$("selector").removeClass("className")

05.How do you toggle a class?

```
$("selector").toggleClass("className")
```

06.How do you append content to an element?

```
$("selector").append("<p>Appended</p>")
```

07. How do you prepend content to an element?

```
$("selector").prepend("<p>Prepended</p>")
```

08.How do you remove an element from the DOM?

```
$("selector").remove()
```

09.How do you clone an element?

```
$("selector").clone()
```

Effects & Animations

01. How do you hide an element?

```
$("selector").hide()
```

02.How do you show an element?

```
$("selector").show()
```

03.What does `.fadeIn()` do?

Fades in the selected elements.

04.What does `.fadeOut()` do?

Fades out the selected elements.

05.How do you animate CSS properties?

Using `.animate()` method.